



Greedy Problem Solving 4

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Problem 1

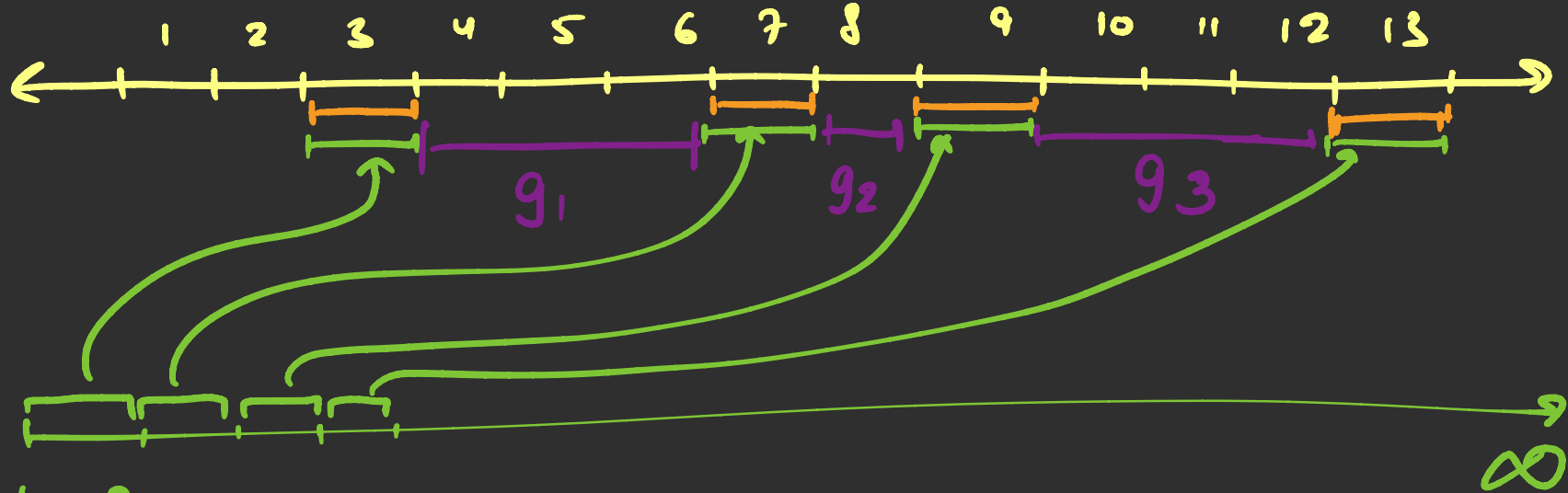


Tape:

<https://codeforces.com/problemset/problem/1110/B>

4 broken segment

gaps $\rightarrow$



$$k = 2$$

$$\underline{p = 4} \rightarrow 4 \leq 2$$

$$L = 4$$

$$p = 4 \rightarrow k$$

$$\underline{\underline{n - k}}$$

$g_1 \quad g_2 \quad g_3 \quad \dots$

Problem 2



Minimize The Error:

<https://codeforces.com/problemset/problem/960/B>

$$a = a_1 \quad a_2 \quad a_3 \quad a_4 \quad \dots \quad a_n$$

$$b = b_1 \quad b_2 \quad b_3 \quad b_4 \quad \dots \quad b_n$$

$$\underline{\xi} = (a_1 - b_1)^2 + \dots + (a_n - b_n)^2 \quad \downarrow$$

lowest

$$k_1 \rightarrow a$$

$$k_2 \rightarrow b$$

$$(a_1 - b_1)^2$$

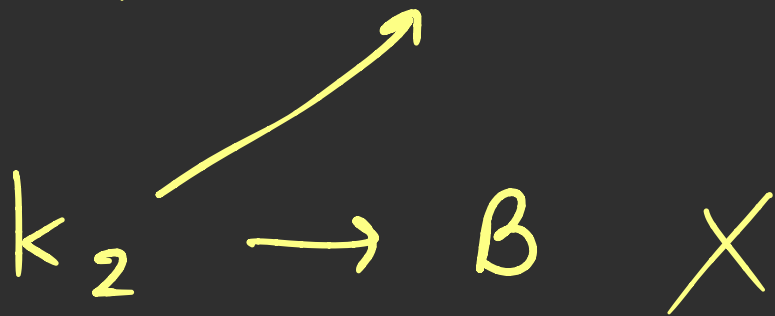
$$a_1 = 6 \rightarrow 5$$

$$b_1 = 3 \rightarrow 4$$

$$\text{diff} = 3$$

$$\underline{(5, 3)}$$

$$\underline{(6, 4)}$$



$$\underline{\underline{k = k_1 + k_2}} \rightarrow \text{I start } \underline{\underline{\text{ops}}}$$

on an arr A st.

$$E = (a_1 - b_1)^2 \dots (a_n - b_n)^2 \downarrow$$

$$\{a_1 \dots a_n\}$$

$$\xi = d^2$$

$$\xi' = (d-1)^2$$

$$\Delta = \xi - \xi'$$

$$= d^2 - (d-1)^2$$

$$= d^2 - (d^2 + 1 - 2d)$$

$$\Rightarrow (2d - 1)$$

max-heap



differences

→ (top)



(LD-1)

pop
←

LD

$k_1 + k_2$

max-heap



$$d_1^2 + d_2^2 + d_3^2 + \dots + d_n^2$$

$$\underline{\xi = \sum d^2}$$

ans

Problem 3



Recover an RBS:

<https://codeforces.com/problemset/problem/1709/C>

(9[↓]))
(

YES

(()) → RBS

(9 9)

NO

(())

() ()

((() ()))

'('s = 2 4 → I have to make
2 more '('s

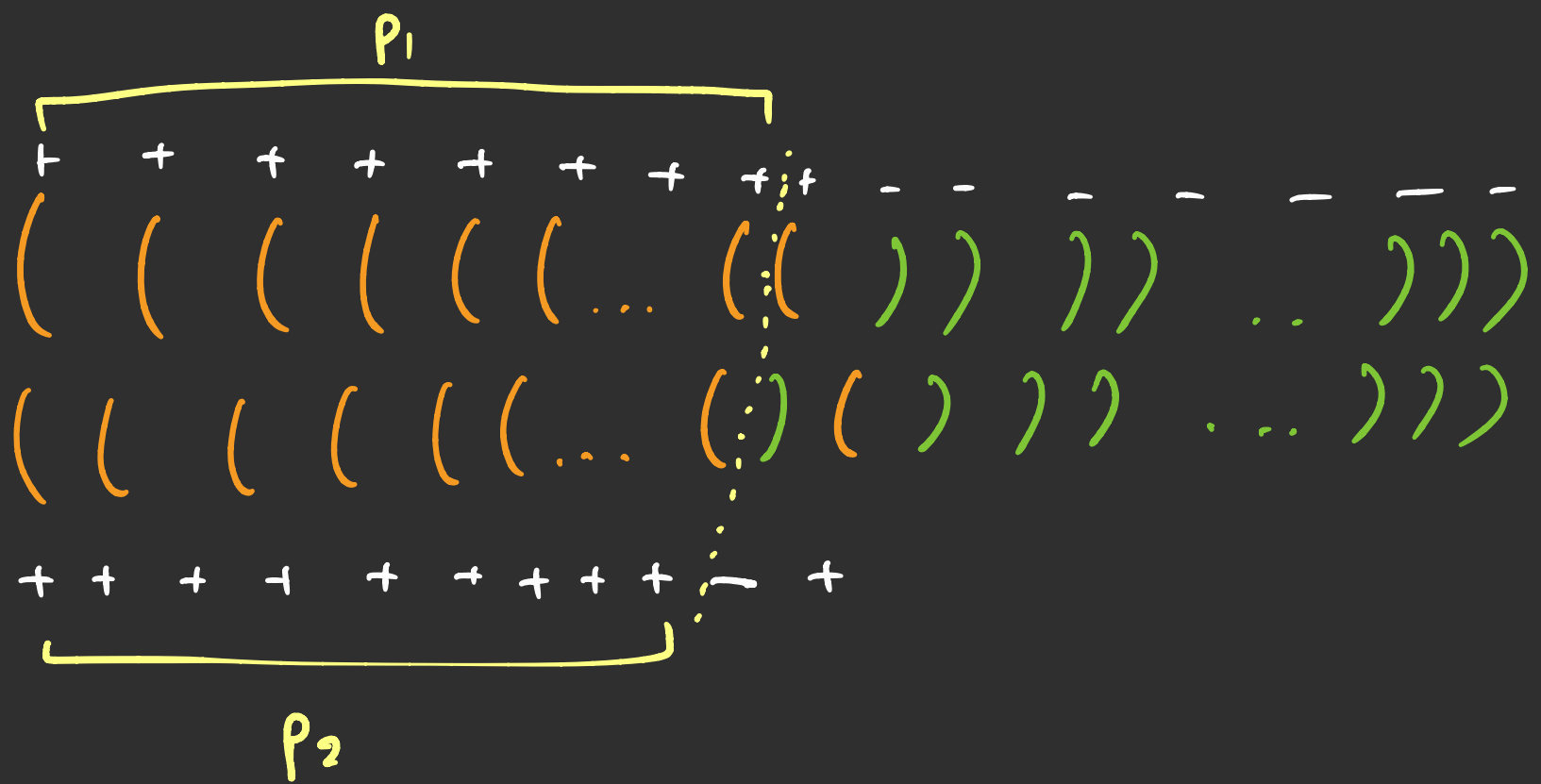
)'s = 3 4 → I have to make
1 more ')'s

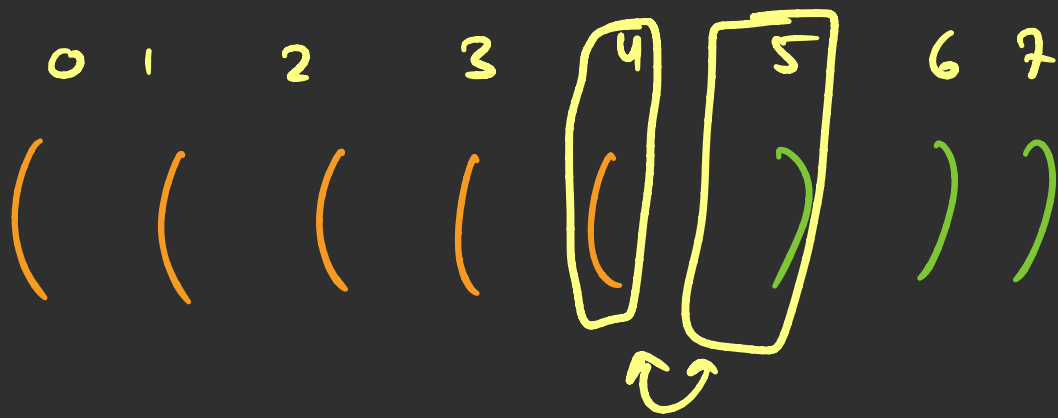
$$n = 8/2 = 4$$

((())))

→ RBS

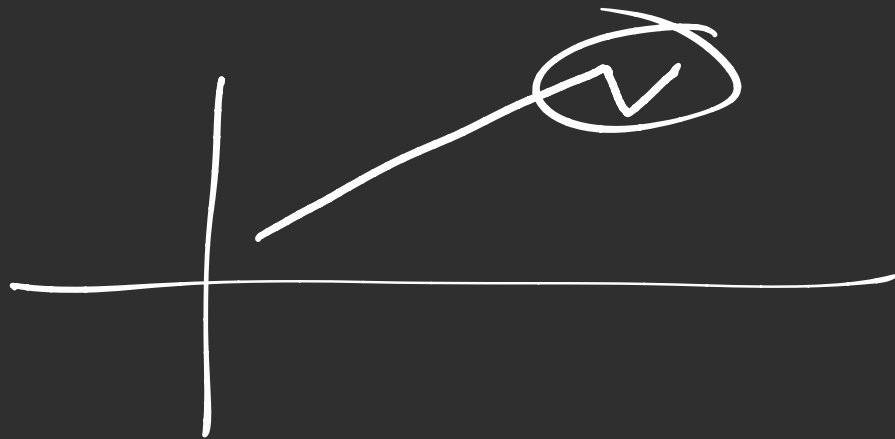
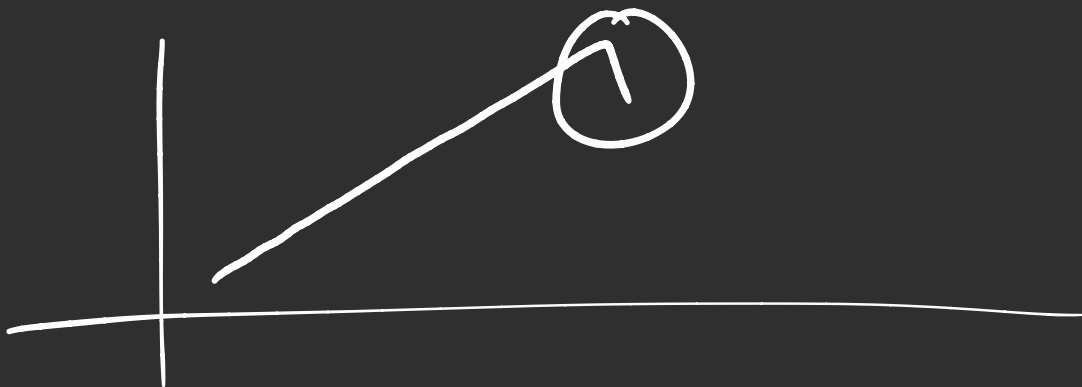
(()





$$op = 5 - 1 = 4$$

$$cl = 3$$



RBS $\rightarrow$ +1 -1 +1

① $('s = ')s$

② at any given moment

$('s \geq ')s$

